
Plant Load Radar

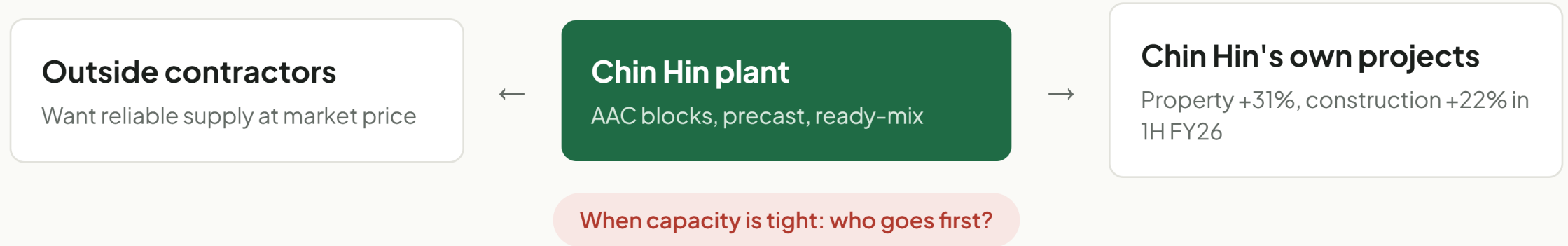
Who gets the capacity when there isn't enough, decided by what's at stake in ringgit

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Two fair priorities, and no shared rule



Today, it's decided deal by deal

The plant chases volume and margin. The group wants its projects on schedule. Both are reasonable.

And planning is blind

Each plant forecasts alone in spreadsheets: excess stock in one place, missed orders in another.

The brief rules out a new ERP, a consultant, a transfer-pricing policy, and escalating upwards.

Plant Load Radar: one rule, one owner, one board

Every week, the plant knows who gets the capacity, and why, in ringgit.

1

The rule

Serve the request with more ringgit at stake: a project's cost of delay, or an order's margin.

2

The owner

The plant scheduler applies it every week. Exceptions go to a 30-minute huddle, never upwards.

3

The board

One view of all committed demand against capacity, built in Excel and Power Apps.

An agentic AI layer does the legwork: reading orders, testing options and drafting messages. People make the call.

More ringgit at stake wins

Example: week 43 is 2,000 m³ short

Project A (Chin Hin's own)

RM82,000 a day

if its handover slips past the deadline

Contractor B (outside)

RM18,000

margin on this order

Serve Project A. Offer Contractor B the next free week.

Keeping it fair

Plan ahead, get protected

Projects that confirm call-offs early get reserved capacity. Unused slots are released.

Promises are kept

Confirmed orders are never bumped. The rule only decides new requests.

Close call? First confirmed wins

If both sides are within about 10%, the earlier confirmed request goes first.

Fill the quiet weeks

AAC can be stocked, so quiet weeks pre-build for busy ones, within a stock limit.

Who decides, and when

1

Project planners and sales

Confirm orders and call-offs early

As they come in

2

Plant scheduler

Updates the board and applies the rule

Every week

3

Plant manager and project director

Settle anything the rule can't, in a 30-minute huddle

Every week

4

Group finance

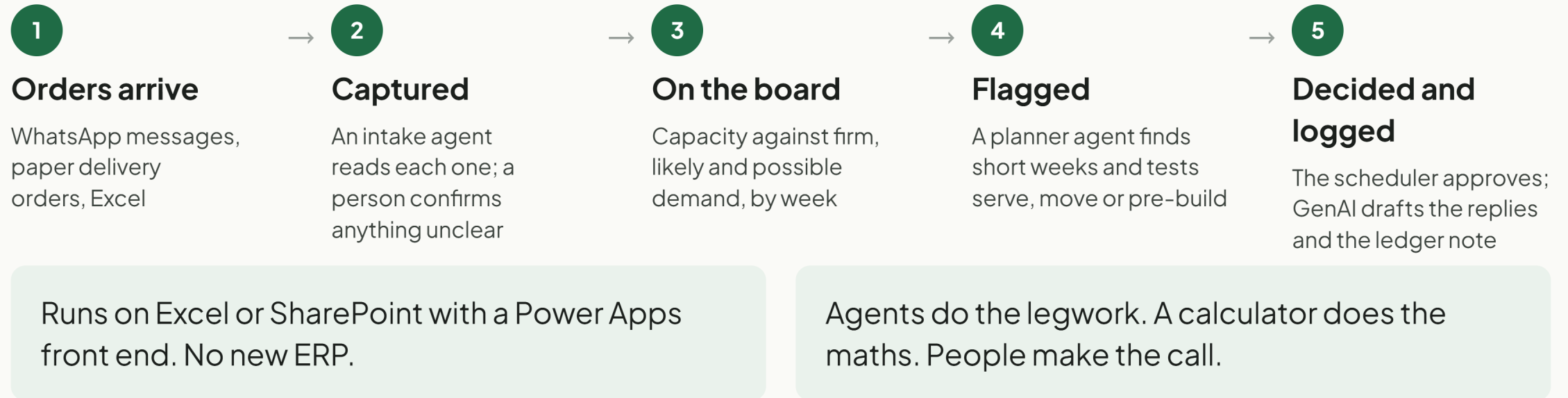
Reviews the ringgit saved and lost

Every month

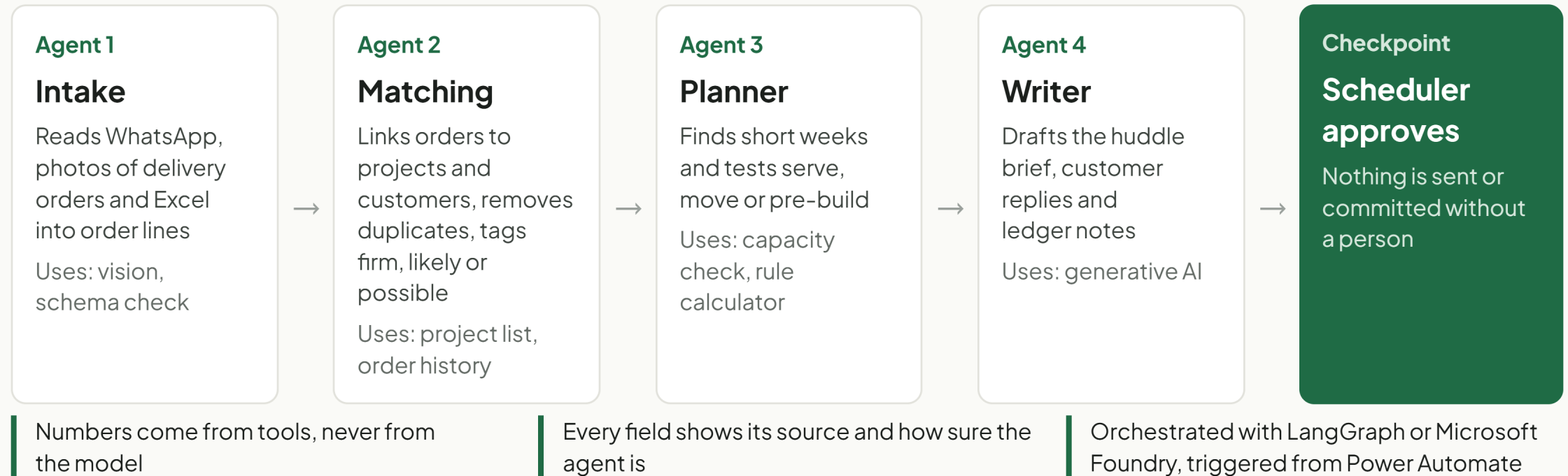
Nothing goes up the chain. The rule settles it.

The plant manager owns every decision made within the rule.

How Plant Load Radar works, every week



Four agents, one human checkpoint



Still premature: forecasting models and letting AI decide. Both come later, once the data is clean.

Intake agent up close

From a WhatsApp message to a clean order

What the board receives

What actually arrives

Boss, Project A site need 1,500 m3 AAC 100mm by Tue 20 Oct. Can confirm ya

9:42 AM

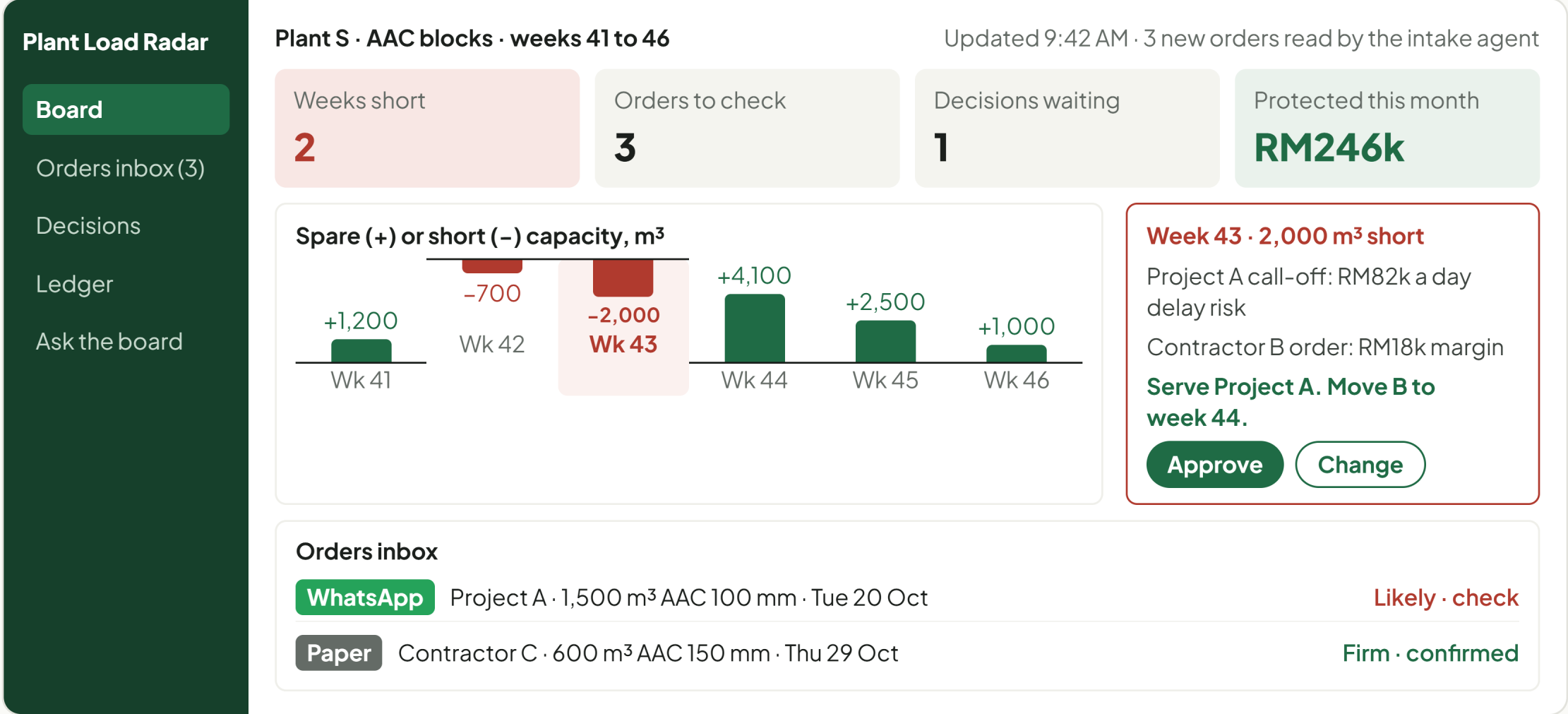


Customer	Project A (internal)	Confirmed
Product	AAC block, 100 mm	Confirmed
Volume	1,500 m³	Confirmed
Needed by	Tue 20 Oct	Confirmed
Firm or likely?	Likely: no call-off yet	Check

If the AI isn't sure, it asks a person. It never guesses.

Illustrative message. Same source-first approach as VERA AI, the knowledge assistant I built at PETRONAS.

The dashboard: Plant Load Radar in one screen



Writer agent output

Generative AI does the writing, people send it

Huddle brief, drafted for the plant manager

Plant S · this week's decisions

Week 43 is 2,000 m³ short. Recommended: serve Project A (about RM82,000 a day of delay risk) and move Contractor B to week 44.

Week 42 is 700 m³ short. Covered by AAC pre-built in week 41.

Needs your call: one decision, week 43.

Reply to Contractor B, waiting for approval

Hi, thanks for your order of 1,500 m³ AAC 100 mm. Week 43 is fully booked, but we can deliver from Mon 26 Oct. Shall I lock that in for you?

Approve and send

Edit

Ask the board

Scheduler: "Can we take 800 m³ for Johor next week?" **Agent:** "Yes in week 44, which has 4,100 m³ free. Week 43 would go 2,800 m³ short."

Illustrative outputs. The numbers come from the rule calculator; the model only writes the words.

Rolling it out: about three months at one plant

Month 1

Understand and build

- Sit with the plant scheduler
- Rebuild recent decisions from WhatsApp and paper, and price them
- Build the dashboard and the agents

By the end: a working dashboard with the plant's real orders

Month 2

Agree and go live

- Agree the rule with the plant manager and a project director
- Start the weekly 30-minute huddle
- Log every decision with its ringgit

By the end: the plant decides by the rule every week

Month 3

Measure and decide

- Compare cash, margin and delays with before
- Fix what didn't work
- Decide whether to roll out to precast and ready-mix

By the end: a ringgit result and a go or no-go

It worked if we see

Cash released from excess stock

Outside margin no longer lost

Fewer project delay days

About me

Fakhrul Azim · Final-year Computer Science, Data Analytics, Universiti Teknologi PETRONAS

I build agentic and generative AI that people actually use, and I can show the working.

Achievements

Dean's List, three semesters

Universiti Teknologi PETRONAS

IEEE paper accepted, MCAIT 2026

Graph-grounded generative interfaces; presenting in Madinah

HACKaSTONE 2026 Grand Final

Qualified for Amsterdam with Team EduNova

Codex Community Day presenter

Presented Synapse with Team EduNova

Esports gold, twice

SUKIPT and Hotlink MLBB Unilegends; team went on to represent Malaysia in Singapore, 2024

Projects and experience

VERA AI · PETRONAS internship

RAG knowledge assistant with strong benchmark results

Technical Lead, System Automations

UTP Career Development Office: 14 undocumented Power Automate flows rebuilt

Arcana · final-year project

Multi-agent pipeline on LangGraph, GraphRAG and Neo4j

Synapse · founder, PocketLab

Adaptive, agentic science learning for SPM students

OmniX and Axiom · side projects

Local-first AI creative suite; generative-UI learning platform

Leadership

Project Director, Griffin Esports Club

Led UTP's esports club and ran a 16-team tournament

LEADX 2.0 Symposium

Sponsorship and Collaboration committee

GDSC-UTP

Ceremony and Protocol

Korea outreach programme, Seoul

Coordinated cultural performances and a food showcase

Plant Load Radar · a proposal for Chin Hin Group

Thank you

One rule, one owner, one board, and agents doing the legwork.

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